

CHARGE FORWARD: High-Voltage Batteries And MSMEs Can Fuel INDIA'S EV REVOLUTION

A nuanced explanation of low- and high-voltage EV batteries by Dr Gokhale, Vice President for Battery Technology at JSW Energy, illuminates their advancements and influence. The essential role of MSMEs and academia in developing a strong EV ecosystem emerges, marking their significance for India's push towards self-reliance in this innovative field.

Dr Gokhale, can you explain the difference between low-speed EV batteries and high-speed, high-voltage EV batteries, perhaps with an example?

Typically, low-voltage batteries are defined as those up to 48 volts, while high-voltage batteries exceed this threshold. This distinction is fundamental in electrical engineering studies. The critical difference lies in the electron transport between the anode and cathode during energy generation, which can be stored in various forms, such as batteries or supercapacitors.

Are most vehicles in India equipped with low- or high-voltage systems?

Most vehicles in India, particularly two-wheelers and three-wheeler autos, use low-voltage systems around 48 volts. Brands like Ola, Ampere, Bgauss, and Bajaj predominantly use these low-voltage batteries.

Which type of battery do you consider safer for electric vehicles?

Both types have their risks if not



DR YASHODHAN P. GOKHALE
VICE PRESIDENT FOR BATTERY TECHNOLOGY, JSW ENERGY